ENGR 213 | Ordinary Differential Equations

Recovered ODE exercises from your earlier chats and ENGR 213 tutor notes. Every question is followed by small, beginner-friendly steps. Questions whose screenshot wording could not be verified are excluded. Not a complete archive of every screenshot.

1. Substitution (recent tutorial)

QUESTION: $y' = (3x + y)^2 - 3$, with $y(0) = 1$.

Step 1. Notice the repeated expression 3x + y. Call it u.

```
u = 3x + y
```

Step 2. Differentiate BOTH sides with respect to x. The derivative of y is y'.

```
u' = 3 + y'
```

Step 3. Replace y' using the given equation.

```
u' = 3 + [(3x+y)^2 - 3] = u^2
```

Step 4. Move u terms to the left and x terms to the right.

```
du/u^2 = dx
```

Step 5. Integrate. The power rule gives integral u^(-2) = -u^(-1).

```
-1/u = x + C
```

Step 6. Use y(0)=1. At x=0, u=3(0)+1=1.

```
-1/1 = 0 + C => C = -1
```

Step 7. Solve for u and then replace u with 3x+y.

```
-1/u = x-1 => u = 1/(1-x)
```

FINAL ANSWER: y = 1/(1 - x) - 3x (on the interval containing x=0).

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2. Bernoulli (recent tutorial)

QUESTION: $y' - x^4y = (3/5)x^4y^{-4}$.

Step 1. Bernoulli form is $y' + P(x)y = Q(x)y^n$. Here n=-4.

```
n = -4, so 1-n = 5
```

Step 2. Set $u=y^{1-n}$, so $u=y^5$. Differentiate using the chain rule.

```
u = y^5 => u' = 5y^4 y'
```

Step 3. Multiply the ORIGINAL equation by $5y^4$.

```
5y^4 y' - 5x^4 y^5 = 3x^4
```

Step 4. Substitute u' and u. Now the equation is LINEAR.

```
u' - 5x^4 u = 3x^4
```

Step 5. Find the integrating factor: $P(x)=-5x^4$.

```
IF = exp(integral -5x^4 dx) = exp(-x^5)
```

Step 6. Multiply through by the integrating factor and recognize a product derivative.

```
(u exp(-x^5))' = 3x^4 exp(-x^5)
```

Step 7. Integrate the right side using $t=-x^5$, $dt=-5x^4dx$.

```
u exp(-x^5) = -(3/5)exp(-x^5) + C
```

Step 8. Multiply by $\exp(x^5)$ and replace $u=y^5$.

```
y^5 = C exp(x^5) - 3/5
```

FINAL ANSWER: $y = [C \exp(x^5) - 3/5]^{1/5}$ (real fifth root).

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3. Separable equation with initial condition

QUESTION: $y' = (x - 3)e^{-2y}$, with $y(3)=\ln(3)$.

Step 1. Write $y'=dy/dx$ and move e^{-2y} to the left.

```
exp(2y) dy = (x-3) dx
```

Step 2. Integrate both sides. On the left, exp(2y) integrates to exp(2y)/2.

```
exp(2y)/2 = (x-3)^2/2 + C
```

Step 3. Multiply all terms by 2 and rename the constant.

```
exp(2y) = (x-3)^2 + C
```

Step 4. Plug in x=3 and y=ln 3. Remember exp(2 ln3)=9.

```
9 = 0 + C => C = 9
```

Step 5. Take ln of both sides, then divide by 2.

```
2y = ln((x-3)^2 + 9)
```

FINAL ANSWER: $y = (1/2) \ln[(x-3)^2 + 9]$.

---

4. Constant solution (initial value problem)

QUESTION: $y' = y^2 - 4y + 3$, with $y(0)=3$.

Step 1. Factor the right side.

```
y' = (y-1)(y-3)
```

Step 2. Try the constant value given in the initial condition: y=3.

```
y' = 0 and (3-1)(3-3)=0
```

Step 3. Both sides are 0, so y=3 solves the differential equation. It also satisfies y(0)=3.

```
y(x) = 3
```

Reminder. Do not divide by y-3 first: that would throw away this valid constant solution.

FINAL ANSWER: y(x) = 3.

---

5. Linear ODE: integrating factor (from tutor notes)

QUESTION: y' + (1/x)y = x^2 - 1, x > 0.

Step 1. This matches y' + P(x)y=Q(x), with P(x)=1/x.

```
P(x) = 1/x
```

Step 2. Find integrating factor IF=exp(integral P(x) dx). Since x>0:

```
IF = exp(ln x) = x
```

Step 3. Multiply EVERY term by x.

```
xy' + y = x^3 - x
```

Step 4. Recognize the product rule backwards: (xy)'=xy'+y.

```
(xy)' = x^3 - x
```

Step 5. Integrate both sides.

```
xy = x^4/4 - x^2/2 + C
```

Step 6. Divide each term by x.

```
y = x^3/4 - x/2 + C/x
```

FINAL ANSWER: y = x^3/4 - x/2 + C/x, x > 0.

---

6. Homogeneous substitution (from tutor notes)

QUESTION: (x - y)dx + xdy = 0, x not equal to 0.

Step 1. Write it in derivative form by dividing by dx.

```
x-y + xy' = 0
```

Step 2. Rearrange: y' depends on y/x, so substitute y=ux.

```
y' = y/x - 1; y = ux
```

Step 3. Use the product rule to differentiate y=ux.

```
y' = u + xu'
```

Step 4. Replace y/x with u and y' with u+xu'.

```
u + xu' = u - 1
```

Step 5. Cancel u on both sides.

```
xu' = -1 => du = -(1/x)dx
```

Step 6. Integrate.

```
u = -ln|x| + C
```

Step 7. Replace u=y/x, then multiply by x.

```
y/x = C - ln|x|
```

FINAL ANSWER: y = x[C - ln|x|].

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7. Type-1 substitution (from tutor notes)

QUESTION: $y' = (x + y + 1)^2$.

Step 1. The expression x+y+1 repeats. Let u equal it.

```
u = x+y+1
```

Step 2. Differentiate both sides.

```
u' = 1+y'
```

Step 3. Replace y' with u squared.

```
u' = 1+u^2
```

Step 4. Separate variables.

```
du/(1+u^2) = dx
```

Step 5. Integrate; the antiderivative of 1/(1+u squared) is arctan(u).

```
arctan(u) = x+C
```

Step 6. Apply tangent on both sides.

```
u = tan(x+C)
```

Step 7. Replace u=x+y+1 and solve for y.

```
x+y+1 = tan(x+C)
```

FINAL ANSWER: y = tan(x+C) - x - 1 (where tangent is defined).

Study tip: Identify the equation type first. Separable: move x and y apart. Linear: integrating factor. Bernoulli: u=y^(1-n). Homogeneous: y=ux. Repeated ax+by+c: set u=ax+by+c.